

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Fuzzy Logic and Reasoning**

Q2b. Solution:

Rule 1 - Consequent Membership Function:

$$B1 = \text{TrapMF}(y, [0, 2, 4, 6])$$

$$\text{Centroid: } (6 - 0) / 2 = 3$$

$$\text{Area: } [(2 + 6) / 2] * 1 = 4$$

Rule 2 - Consequent Membership Function:

$$B2 = \text{TriMF}(y, [4, 6, 8])$$

$$\text{Centroid: } (4 + 8) / 2 = 6$$

$$\text{Area: } (8 - 4) / 2 = 2$$

Rule 3 - Consequent Membership Function:

$$B3 = \text{TrapMF}(y, [6, 8, 10, 12])$$

$$\text{Centroid: } (6 + 12) / 2 = 9$$

$$\text{Area: } [((12 - 6) + (10 - 8)) / 2] * 1 = 4$$

$$Y = \frac{(0.2 * 4 * 3) + (0.2 * 2 * 6) + (0.6 * 4 * 9)}{(0.2 * 4) + (0.2 * 2) + (0.6 * 4)}$$

$$Y = \frac{26.4}{3.6}$$

$$Y = 7.3333$$